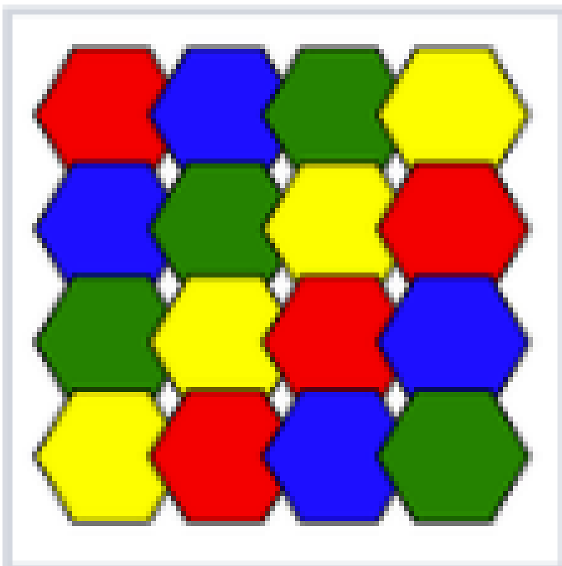
LM Studio

Model: qwen/qwen3.8-27b

Write a turtle progr

User

Write a turtle program to recreate this. Look at the output and iterate until you no longer are making progress.



Assistant

```
import turtle

def draw_hexagon(t, x, y, size, color):
    """
    Draws a flat-top regular hexagon centered at (x, y) with a given side length.
    """
    t.penup()
    # Move to the rightmost vertex to start drawing
    t.goto(x + size, y)
    t.pendown()
    # Set heading to the top-right vertex (120 degrees from positive x-axis)
    t.setheading(120)

    t.fillcolor(color)
    t.pencolor('black')
    t.begin_fill()
    for _ in range(6):
        t.forward(size)
        t.left(60)
```

```python
    t.end_fill()

def main():
    screen = turtle.Screen()
    screen.bgcolor('white')
    t = turtle.Turtle()
    t.speed(0)  # Fastest drawing speed
    t.hideturtle()

    # 1. Draw the red border
    border_size = 300
    t.penup()
    t.goto(-border_size/2, -border_size/2)
    t.pendown()
    t.pencolor('red')
    t.pensize(2)
    for _ in range(4):
        t.forward(border_size)
        t.left(90)

    # 2. Define the grid colors (Row by Row, Left to Right)
    colors = [
        ['red', 'blue', 'green', 'yellow'],
        ['blue', 'green', 'yellow', 'red'],
        ['green', 'yellow', 'red', 'blue'],
        ['yellow', 'red', 'blue', 'green']
    ]

    # 3. Grid parameters
    size = 45      # Side length of the hexagon
    dx = 60        # Horizontal spacing between centers
    dy = 60        # Vertical spacing between centers
    start_x = -90  # X coordinate of the first column center
    start_y = 90   # Y coordinate of the first row center (top)

    # Set line thickness for the hexagon outlines
    t.pensize(3)

    # 4. Draw the hexagons
    # Drawing row-by-row, left-to-right ensures that lower/right hexagons
    # correctly overlap the upper/left ones, matching the image.
    for r in range(4):
        for c in range(4):
            x = start_x + c * dx
            y = start_y - r * dy
            draw_hexagon(t, x, y, size, colors[r][c])

    turtle.done()

if __name__ == '__main__':
    main()
```

User

continue